

Telecom Provider Agent

AI in Enterprise - Ollama Prototype Development

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Abstract

This paper explores the feasibility of an AI-based telecom recommendation agent that analyzes user usage patterns and compares them against available tariff options in order to identify potential cost improvements. The system leverages large language models executed either locally or via a cloud-based backend and is designed with a strong emphasis on modularity, interpretability, and reproducibility.

To enable controlled experimentation, the tariff knowledge base is deliberately limited to a small set of plans from two providers, sourced either from a static JSON file or through constrained web scraping. The reasoning agent evaluates overprovisioned and underprovisioned usage scenarios and produces a single, structured recommendation along with an explicit justification.

Experiments compare local and cloud-based model executions under identical prompts and data constraints. The results show that larger cloud-based models consistently provide more accurate and clearer recommendations, particularly in scenarios requiring comparative reasoning across multiple tariff options, while smaller local models exhibit limitations in identifying cost-saving alternatives. Evaluation is performed using manually assigned accuracy and clarity scores, which are logged for analysis.

Overall, the findings indicate that large language models can serve as effective reasoning components for usage-aware telecom cost optimization when applied under controlled and reproducible conditions. The system is intended as a feasibility prototype rather than a production-ready solution and highlights both the potential and current limitations of LLM-driven decision support in the telecom domain.

1. Introduction

The telecommunications industry forms the backbone of global communications, with data consumption expected to increase from 3.4 million petabytes (PB) in 2022 to 9.7 million by 2027 [1]. On one hand, the competitive environment necessitates ongoing improvements in customer service to sustain and increase market share, leading to a critical challenge: enhancing customer service efficiency and accuracy without compromising user privacy [2]. On the other hand, despite widespread commoditization, only 34% of telecommunications customers feel satisfied with their service, and 70% are frustrated by a lack of consistent experience across channels. Only one in three customers stay with their service provider for over five years [3]. One reason for their unsatisfaction is that users frequently overpay due to mismatches between their actual usage and their subscribed plans, but mobile telecom tariffs could be complex and difficult for consumers to compare. Recent advances in Large Language Models (LLMs) enable the creation of intelligent agents capable of reasoning over structured data and providing human readable recommendations.

This project explores the feasibility of a telecom provider agent that uses a local/cloud LLM to analyze users' master and usage data and compare it against available telecom tariffs. The goal is to recommend cost improvements while providing clear reasoning and ensuring reproducibility of the process.

The main contributions of this work are:

- A reproducible LLM-based telecom optimization agent
- A rule-constrained prompting strategy to ensure deterministic and verifiable outputs
- An experimental framework with logging of quality-related criteria such as response time, clarity and accuracy

2. System Design

The system consists of three main components:

- **User Data Input** captures the user's current plan details and actual monthly usage, which serves as the primary input to the system.
- **Tariff Knowledge Base** stores available telecom offers, either loaded from a local JSON file or obtained via controlled web scraping. This component ensures reproducibility of tariff data.
- **LLM-Based Reasoning Agent** compares user data against tariffs and produces a structured recommendation.

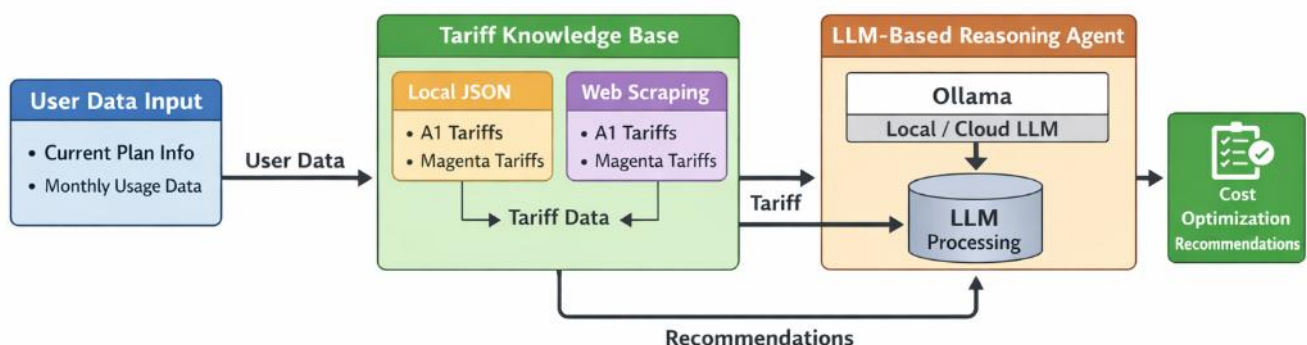


Figure 1: System architecture of the proposed AI-based telecom recommendation agent. The figure was generated with the assistance of ChatGPT.

The overall workflow starts with collecting user data, followed by tariff retrieval. The processed information is then passed to the LLM-based agent, which performs reasoning and generates cost-saving recommendations.

The project is implemented in Python (≥ 3.12) and uses Ollama to run a large language model locally with optional support for cloud-based models.

2.1 User Data Input

User data includes:

- Current monthly price (€)
- Included data (GB, 0 means unlimited)

- Included minutes
- Included SMS
- Actual data used (GB)
- Actual minutes used
- Actual SMS used

2.2 Tariff Knowledge Base

The goal of this project is to explore the feasibility of an AI-based telecom agent in both local and online settings, rather than to build a comprehensive commercial tariff comparison system. Therefore, we intentionally avoid relying on large-scale third-party websites that aggregate tariffs from all providers.

Using such sources would require maintaining a large and frequently changing set of tariffs in the local JSON knowledge base, making it difficult to ensure consistency and comparability between local and web-based data sources. This would in turn hinder a clear evaluation of whether the agent produces consistent recommendations when switching between local and online tariff inputs.

To keep the comparison transparent and reproducible, we restrict the web-based tariff source to two providers: A1 (<https://www.a1.net/handys-tarife/tarife-ohne-bindung>) and Magenta (<https://shop.magenta.at/handytarife-ohne-smartphone>). Correspondingly, the local JSON knowledge base contains a curated subset of tariffs selected from these two providers, representing small, medium, and large plans. This design allows the results from JSON-based and web-based inputs to be compared in a clear and controlled manner.

2.2.1 Local JSON file

The static *tariffs.json* file contains 6 structured tariffs. Three of them are from A1, the others are from Magenta. They were given in the following structure, for example:

```
{  
  "provider": "A1",  
  "name": "A1 SIMply L",  
  "data_gb": "unlimited",  
  "minutes": "unlimited",  
  "sms": "unlimited",  
  "price": 29.9,  
  "url": "https://www.a1.net/handys-tarife/tarife-ohne-bindung"  
}
```

2.2.2 Internet source

The two internet sources were given above. Tariff data is collected from them using a lightweight scraping approach. Raw HTML content is downloaded and parsed to remove non-informative elements such as scripts and styles. The remaining text is extracted and truncated to a fixed size to control prompt length. In fallback scenarios, a character-level truncation of the raw HTML is applied. This design ensures robustness and reproducibility while remaining sufficient for comparing local and web-based tariff inputs.

2.3 LLM-Based Reasoning Agent

2.3.1 Ollama Integration and Model Selection

Ollama is used in this project as the local inference engine. The agent communicates with the models using the Ollama Python client, ensuring that all inference runs locally and can be reproduced on any machine with sufficient resources.

The reasoning component of the system is implemented using large language models executed either locally or via a cloud-based backend through the Ollama framework. Specifically, we use deepseek-v3.1:671b accessed through the Ollama API for cloud-based inference, and qwen2.5:7b for local execution.

These models were selected due to their strong reasoning capabilities and native support within the Ollama ecosystem, which simplifies integration and deployment. In particular, the cloud-based deepseek-v3.1 model offers significantly lower inference latency compared to locally installed large models, making it suitable for interactive experimentation. The smaller qwen2.5 model enables local execution, supporting reproducibility and offline evaluation.

The system design allows the reasoning component to be easily replaced with other Ollama-compatible models, facilitating comparative experiments and future extensions without changes to the overall pipeline.

2.3.2 Prompting Strategy

All inputs to the LLM are explicitly structured to reduce ambiguity and improve consistency across runs. The prompt is composed of three main sections: current subscription details, user usage data and a list of candidate tariffs obtained from either the local JSON knowledge base or online sources.

Each section is clearly delimited and labeled, ensuring that the model can reliably distinguish between user-specific information and tariff descriptions. The prompting strategy emphasizes comparison rather than free-form generation by instructing the model to evaluate each candidate tariff relative to the user's observed usage. Additionally, the prompt includes explicit instructions to focus on cost implications and to avoid introducing assumptions beyond the provided data.

2.3.3 Reasoning and Output Structure

The LLM is instructed to perform explicit reasoning steps before producing a final recommendation. Instead of generating unstructured text, the model is guided to produce a structured output that separates the recommendation from its justification.

The output includes:

- Current plan summary
- Decision (keep or change plan)
- Suggested new plan (if applicable)
- Offer link
- Estimated savings
- One-sentence reasoning

This structured format improves transparency and allows the results to be compared across different input sources.

2.3.4 Evaluation Criteria

The system outputs were evaluated using two qualitative criteria: accuracy and clarity. Both metrics were assigned manually by the authors after inspecting the model-generated recommendations.

Accuracy (1–5) measures whether the recommended decision (e.g., keeping the current plan or switching to a new tariff) is economically reasonable given the user's actual usage and the available tariff options. A high accuracy score indicates that the recommendation correctly identifies overprovisioned or underprovisioned plans and suggests an appropriate alternative when applicable.

Clarity (1–5) evaluates the quality of the explanation accompanying the recommendation. This includes the coherence of the reasoning, explicit reference to usage and pricing data and the overall comprehensibility of the output for a non-expert user.

The corresponding model outputs (see 2.3.3), response time (seconds) and assigned scores were stored in a structured log for analysis. Although these evaluations were not used for automated model adaptation in the current implementation, they provide a basis for future work on feedback-driven improvement.

3. Experiments and Results

3.1 Experimental Setup

We evaluate the proposed reasoning agent using a set of representative usage scenarios that reflect common telecom subscription patterns. The experiments vary across three dimensions: model

execution setting (local vs. cloud), tariff source (local JSON knowledge base vs. web-based tariffs) and user usage profiles, including overprovisioned, underprovisioned and already optimal plans.

For each experiment, the model-generated recommendation was reviewed manually and assigned accuracy and clarity scores on a five-point scale. These scores reflect human judgment rather than automated metrics and were used solely for comparative analysis across different models and data sources.

3.2 Results and Analysis

The experimental results reveal a clear performance gap between the locally executed model and the cloud-based model. In particular, the local qwen2.5 model frequently failed to identify cost-saving opportunities in overprovisioned scenarios. In such cases, the model tended to recommend keeping the current plan, even when more suitable and cheaper alternatives were present in the tariff knowledge base.

In contrast, the cloud-based model (deepseek-v3.1:671b) consistently produces accurate recommendations across tested scenarios. It reliably identifies overprovisioned plans and suggests lower-cost alternatives, while also recognizing cases where upgrading to a higher-tier plan reduces overage costs. Inference times remain stable for a few seconds, making the model suitable for interactive use.

When comparing tariff sources, the agent shows consistent reasoning behavior across local JSON and web-based inputs. Although the exact recommended tariff may differ depending on availability (e.g., A1 versus Magenta), the overall decision logic and estimated cost savings remain comparable. This demonstrates that the agent's reasoning is driven by usage patterns rather than by specific data representations.

Overall, the agent did not hallucinate tariffs or prices and kept a fixed output format to simplify evaluation and comparison. The experiments confirm that the proposed agent can provide plausible, transparent and stable recommendations under controlled conditions, supporting the feasibility of online deployment.

4. Conclusion

This work explored the feasibility of an AI-based telecom recommendation agent that compares user usage patterns against available tariff options using large language models. The system was designed with an emphasis on modularity, interpretability, and reproducibility, enabling controlled comparisons between local and cloud-based execution as well as between static and web-derived tariff sources.

The experimental results demonstrate that large language models can effectively identify economically reasonable tariff recommendations when sufficient reasoning capacity is available. In particular, the cloud-based model consistently detected overprovisioned and underprovisioned plans and produced clear, well-structured explanations. In contrast, the locally executed model showed limitations in comparative reasoning, especially in scenarios requiring optimization across multiple tariff options.

Overall, the findings indicate that LLM-based agents are a viable approach for usage-aware telecom cost optimization, provided that careful constraints are applied to data sources and evaluation procedures. While the system is not intended as a production-ready solution, it serves as a reproducible prototype that highlights both the potential and the current limitations of LLM-driven decision support in this domain.

5. Limitations and Future Work

While the proposed AI-based telecom recommendation agent demonstrates promising results, several limitations remain. First, the tariff knowledge base is intentionally restricted to a small subset of plans from two providers. Although this design choice supports controlled comparison and reproducibility, it limits the coverage of the system and does not reflect the full complexity of real-world telecom markets.

Second, the evaluation relies on manually assigned accuracy and clarity scores. Although this qualitative assessment provides valuable insight into model behavior and reasoning quality, it remains subjective and does not scale to large datasets. Furthermore, the collected evaluation scores were not used for automated model adaptation or feedback-driven learning in the current implementation.

Third, the reasoning agent operates in a single turn setting and does not model user preferences beyond observed usage patterns. Factors such as contract duration, network quality, the actual costs by underprovisioned plan or personal preferences are not considered. Additionally, although output variability is reduced through conservative generation settings, complete determinism cannot be guaranteed when using autoregressive language models like deepseek-v3.1 and qwen2.5.

Future work will focus on expanding the tariff dataset in a controlled manner while introducing versioning mechanisms to preserve reproducibility. Incorporating multi-turn interactions would allow the agent to elicit user preferences and refine recommendations.

Finally, an important direction is to leverage the manually collected evaluation scores as feedback signals, enabling iterative improvement of prompting strategies or model selection in a semi-automated evaluation loop.

6. Reproducibility

The project is reproducible:

1. Clone the public GitHub repository https://github.com/almoualem/ai_telecom_agent
2. Download and Install Ollama <https://ollama.com/download>
3. Install Python >= 3.12 <https://www.python.org/downloads/>
4. Pull the required model “qwen2.5:7b” using Ollama.
5. Create the API key for the cloud model in Ollama: <https://ollama.com/settings/keys>
6. Set up using cmd in the cloned repo folder:
 - `python -m venv venv`
 - `venv\Scripts\activate`
 - `pip install -r requirements.txt`
 - `set OLLAMA_API_KEY=your API key`
 - `echo %OLLAMA_API_KEY%`
7. Run: `python request.py`

References:

- [1] Artificial Intelligence in Telecommunications. White Paper. World Economic Forum 2025.
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